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Spatial Biodiversity Modeling Pipeline and Benchmark

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1 Introduction

1.1 Motivation

Biodiversity is defined as "the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems."[1] Biodiversity, in short, is the collection of all living organisms of an ecosystem from plants to animals to fungi to bacteria to protist. Biodiversity is integral to the functions and stability of ecosystems and plays a pivotal role in maintaining ecosystem functions, recycling nutrients, breaking down pollutants, regulating water systems, carbon storage,stabilizing total biomass within a community, and driving the productivity of an ecosystem (Stork 2009; Cardinale et al. 2012; Gascon et al. 2015). Furthermore biodiversity's significant for human survival is immense, as it provides food, materials, medicine, water security, health, well-being, energy, transportation, climate regulation, and more (Stork 2009; Gascon et al. 2015; Brauman et al. 2020; Kim et al. 2024).

Despite the overwhelming benefits of biodiversity to humanity and ecosystems, biodiversity is declining at an unprecedented rate. Out of all the world's terrestrial wilderness ecosystems approximately 74% of this land area has experienced erosion and approximately 9.6% has been completely lost since the early 1990s (Watson et al. 2016). Some of the most impacted areas of biodiversity loss include marine ecosystems where 66% of the ocean's surface has been significantly impacted by human activities (Díaz et al. 2019). Moreover, it is reasonably likely that most of coral reefs will be lost if global temperatures exceed 2°C (Frieler et al. 2012). Furthermore, human activities threaten the survival of approximately 1 million species, 25% of plants and animals are threatened, and human activities have caused an explosion in the rate of species extinction, which is now estimated to be around 1,000 times the natural background rate (Pimm et al. 2014; Díaz et al. 2019). To make matters worse, conservation efforts are underfunded, with a deficit exceeding half a trillion US dollars annually (Deutz et al. 2020). The acceleration of taxonomic loss and the drastic underfunding of conservation efforts have necessitated a shift towards prioritization of biodiversity conservation and ecological rehabilitation.

Many initiatives around the world are being adopted to combat biodiversity loss and climate change which are intricately linked. These initiatives focus on international cooperation, expanding protected areas, reducing pollution, and incentives for sustainable practices (IPBES 2019; IPCC 2023). Specific policy initiatives such as the 30x30 target-which aims to protect 30% of the world's land area by 2030-represent critical efforts for ecological recovery. If successful, we could see large gains and rebounds in biodiversity. Achieving these goals would have significant impacts including: carbon sequestration, increasing water quality regulation, and expanding capabilities to mitigate nutrient pollution (Zeng, Koh, and Wilcove 2022). For these initiatives to be successful, it is crucial to have accurate and robust models that can predict biodiversity patterns and trends. These models can inform conservation strategies, identify priority areas for protection, and assess the potential impacts of different conservation actions (Sinclair, White, and Newell 2010).

At their core, biodiversity models are spatial models that assume biodiversity is influenced by a wide range of environmental variables at a specific spatial location (Ferrier and Guisan 2006). Over the past few decades, a wide range of models have been developed to predict biodiversity patterns and trends. Broadly, biodiversity models can be grouped into three main families: species distribution models (SDMs), which use a bottom-up or "predict first, assemble later" strategy; joint species distribution models (JSDMs), which jointly model species' responses and co-occurrence patterns through an "assemble and predict together" strategy; and community-level models, which use a top-down approach to model biodiversity metrics or community properties directly (Ferrier and Guisan 2006; Zhang et al. 2019; Pollock et al. 2020). The most common of these models are SDMs, which predict the spatial distribution of individual species based on a wide range of environmental variables (Beery et al. 2021). These individual species predictions can then be aggregated together to predict community level biodiversity metrics (Ferrier and Guisan 2006). Next we have JSDMs, which model multiple species at once using environmental variables and species interactions (Warton et al. 2015). Finally, we have community level models, which model biodiversity as a whole and can predict biodiversity metrics directly without predicting specific species

[1] Convention on Biological Diversity (1993) *Article 2. Use of terms.* Available at: https://www.cbd.int/convention/articles/?a=cbd-02 (Accessed: 8 April 2026).

(Pollock et al. 2020).

While each family has their own strengths and weakness and excels in different context it is unlikely one model will always outperform the others (Ferrier and Guisan 2006; Zhang et al. 2019). But what is clear, is that SDMs are the most researched model family to use for biodiversity prediction. Many studies have focused on improving SDMs, from benchmarks (Valavi et al. 2022) to reviews and evaluations of specific models (Norberg et al. 2019; Waldock et al. 2022), to standardizing best practices for SDMs (Araújo et al. 2019). Similar to SDMs, JSDMs have also been the focus of many studies specifically to combat the neglection of species interactions in SDMs (Wilkinson et al. 2019; Escamilla Molgora et al. 2022). Community level models, on the other hand, are lesser known and underutilized (Nieto-Lugilde et al. 2018). A significant gap remains in our understanding, comparison, and evaluation of SDMs and community-level models, both between these modeling approaches and within the different methods they encompass. This highlights the need for further research and methodological development in community-level biodiversity modeling (Ferrier and Guisan 2006; Nieto-Lugilde et al. 2018; Pollock et al. 2020).

Currently, most community level models are compared within individual studies, across a wide range of datasets, and with varying methodologies, making it difficult to draw general conclusions about their performance and applicability (Cai et al. 2023; Ngor et al. 2023; Baggström et al. 2025). While there have been some reviews of a community level models (Nieto-Lugilde et al. 2018; Mondal and Bhat 2021), there is lack of best practices for this model family which SDMs have (Araújo et al. 2019). This has necessitated the need for a standardized pipeline for community level modeling to allow for more rigorous testing of these models (Pollock et al. 2020). This pipeline would allow for a more systematic evaluation of community level models and provide insights into their strengths and weaknesses in predicting biodiversity patterns and trends. Furthermore, a standardized pipeline would allow for a more nuanced evaluation of community-level models and their applicability across different ecological contexts, including comparisons across taxonomic groups, spatial scales, and environmental predictor sets. This would ultimately contribute to the advancement of biodiversity modeling as a whole and inform conservation strategies and policy decisions.

1.2 Research Objectives and Scope

The primary objective of this master's thesis is to develop a standardized pipeline for community level biodiversity modeling and to provide a case study demonstrating the applicability and flexibility of this pipeline. To achieve this objective, first a pipeline is developed that integrates generalized data preprocessing, feature engineering, model training, and evaluation steps for community level modeling. This pipeline is designed to be flexible and adaptable to a wide range of datasets and modeling approaches. Next, a case study is conducted using this pipeline to compare the performance of different community level models on derived biodiversity metrics from the United Kingdom Butterfly Monitoring Scheme (UKBMS) dataset (UK Butterfly Monitoring Scheme 2026) from the Global Information Biodiversity Facility[2]. Models used in this case study will include statistical models, generalized linear models and generalized additive models, and machine learning models, random forests, gradient boosting, and artificial neural networks (Nelder and Wedderburn 1972; Hastie and Tibshirani 1986; Breiman 2001; Friedman 2001; Lek et al. 1996). These results are then analyzed to provide insights into the strengths and weaknesses of different community level models across different contexts. By applying the pipeline across contexts that vary in taxonomic grouping, spatial scale, environmental predictor sets, and evaluation strategy, this thesis demonstrates its flexibility and general applicability to a range of community-level biodiversity modeling scenarios. This comparative framework also enables a systematic assessment of how different modeling approaches perform under varying analytical conditions, thereby providing insight into their relative strengths, limitations, and appropriate use cases. The key research questions that this master's thesis aims to address are:

1. How do models perform against each other and against an ensemble approach?
2. How do models perform with different strategies of cross validation, specifically random cross validation vs. spatial cross validation?

[2]Global Biodiversity Information Facility (GBIF). Available at: https://www.gbif.org/ (Accessed: 26 April 2026).

3. How do models perform in data limited scenarios?

The main scope of this master's thesis is focused on the development and application of a standardized pipeline for community level biodiversity modeling. While the pipeline is designed to be flexible and adaptable to a wide range of datasets and modeling approaches, the case study conducted in this thesis is limited to the UKBMS dataset and the specific set of modeling approaches described above. The results are therefore intended to illustrate the flexibility and practical utility of the pipeline, while demonstrating how standardized workflows can support more efficient generation of community-level biodiversity insights.

The remainder of this thesis is structured as follows. Section 2 provides background on biodiversity, including how it is defined and measured in Section 2.1, and reviews the main modeling approaches used to predict biodiversity patterns and trends in Section 2.2. Section 3 describes the methodological framework, including the pipeline design in Section 3.1, the case study dataset in Section 3.2, and the models implemented in Section 3.3. Section 4 presents the results of the case study and illustrates the flexibility of the proposed pipeline. Section 5 discusses the implications of these results for biodiversity modeling and conservation, while Section 5.2 addresses relevant ethical considerations. Section 6 outlines potential future directions and applications for research in this area. Finally, Section 7 concludes the thesis by summarising the key findings and their implications for biodiversity modeling and conservation.

2 Background

This section covers biodiversity and how it is measured in Section 2.1 and reviews the model techniques used to predict biodiversity patterns and trends in Section 2.2. This background section provides the necessary context for understanding the methods and results presented in this thesis. It will also highlight the differences between the modeling approaches as well as the different assumptions and limitations of these models.

2.1 Biodiversity and how to measure it

Biodiversity is a complex and multifaceted concept that encompasses the variety of life on Earth, including the diversity of species, genetic diversity within species, and the diversity of ecosystems (Stork 2009). Humanity has relied on biodiversity for survival for thousands of years as it provides us with food, materials, medicine, water security, and has an enormous effect on human health and well-being (Gascon et al. 2015; Naeem et al. 2016; Brauman et al. 2020; Kim et al. 2024). Large losses of biodiversity would be catastrophic for humanity and ecosystems as it would lead to the loss of ecosystem functions, the collapse of food webs, adversely affect human health, degrade medicine production and discovery, and more (Roe, Seddon, and Elliott 2018). The rate of biodiversity loss is unprecedented, and we are quickly moving towards a sixth mass extinction (Cowie, Bouchet, and Fontaine 2022). To combat biodiversity loss effectively, conservation and ecological rehabilitation efforts require accurate, robust models that can predict biodiversity patterns, identify priority areas, and assess potential conservation outcomes (Sinclair, White, and Newell 2010; Pollock et al. 2020).

To effectively predict biodiversity, it is crucial to have a clear understanding of how biodiversity data is collected and measured. Generally, biodiversity data is collected through field surveys, remote sensing, and citizen science initiatives and other schemes depending on what is being measured (Proença et al. 2017; Beery et al. 2021; Brun et al. 2024). These approaches or protocols can provide data on species occurrences, abundances, and distributions across different spatial and temporal scales (Proença et al. 2017). Once this data is collected there are three broad categories of biodiversity metrics that are commonly measured: alpha diversity, beta diversity, and gamma diversity. Alpha diversity refers to diversity on a local spatial scale, such as the diversity within a specific community or ecosystem. Beta diversity refers to the diversity between different communities or ecosystems. Gamma diversity refers to the species diversity across communities or ecosystems within a larger region (Andermann et al. 2022). For the purposes of this thesis, we will focus on alpha diversity, which is the most commonly measured type of biodiversity.

Alpha diversity will be the main focus of this thesis and is one of the most commonly measured types of biodiversity, and represents the diversity within a specific site. Metrics that are included in alpha diversity are species richness, absolute abundance, relative abundance, Shannon index, arithmetic mean abundance, and geometric mean abundance (S. Buckland et al. 2005; S. T. Buckland et al. 2011; Beery et al. 2021). Species richness is the most commonly used metric and simply counts the number of species in a given area (Ferrier and Guisan 2006). Species richness is a very popular metric because presence absence data is the most common type of biodiversity data (Özkan Tümer et al. 2026). Absolute abundance measures the total number of individuals of each species in a given area, and relative abundance measures the proportion of individuals of each species relative to the total number of individuals in a given area (Beery et al. 2021). The Shannon index is another commonly used metric and combines species richness and evenness into a single metric and quantifies the uncertainty in predicting the species of an individual randomly selected from a community (Beery et al. 2021). Arithmetic mean relative abundance is the average abundance of a species in a given area relative to the abundance of the species at a given time. Geometric mean relative abundance is the nth root of the product of the abundances of n species in a given area relative to abundance of that species a specific time (S. Buckland et al. 2005; S. T. Buckland et al. 2011).

For the purpose of this thesis it is beneficial to express these terms mathematically. When calculating biodiversity, we usually look at individual sites. A site is some spatial area that is defined according to a sampling protocol. I define a site as r and for a biodiversity dataset there are n sites which make up the region of interest $R = \{r_1, r_2, \ldots, r_n\}$. Across the entire region of interest R there is a set S of unique

species. Each site r has a set of species S_r which is a subset of the regional species set S.

The alpha diversity metrics used in this thesis can therefore be defined at the site level as follows.

Species richness. Following Beery et al., species richness is the number of unique species in a given site r and can be expressed as:

$$SR_r = |S_r| \,. \tag{1}$$

Total species richness across the entire region of interest R can be expressed as:

$$SR_R = \left| \bigcup_{r=1}^{n} S_r \right| . \tag{2}$$

Absolute abundance. Absolute abundance is the total number of individuals of each species in a given site r and can be expressed as:

$$A_r = \sum_{i \in S_r} a_{i,r} \tag{3}$$

where $a_{i,r}$ is the abundance or count of individuals of species i in site r (Beery et al. 2021).

Relative abundance. Relative abundance is the proportion of individuals of each species relative to the total number of individuals in a given site r and can be expressed as:

$$E_{i,r} = \frac{a_{i,r}}{A_r} \tag{4}$$

Where $a_{i,r}$ is the abundance or count of individuals of species i in site r and A_r is the total abundance of all species in site r (Beery et al. 2021).

Shannon index. The Shannon index combines species richness and evenness into a single metric and quantifies the uncertainty in predicting the species of an individual randomly selected from a community. It can be expressed as:

$$H_r = -\sum_{i \in S_r} p_{i,r} \ln(p_{i,r}) \tag{5}$$

Where $p_{i,r}$ is the proportion of individuals of species i relative to the total number of individuals in a given site r which is the relative abundance $E_{i,r}$. The interpretation of the Shannon index is that higher values of H_r indicate higher diversity (Dušek and Popelková 2017).

While the above metrics focus on the diversity of species in a given site, the following two metrics focus on the change in abundance of species over time. These metrics are useful for measuring changes in biodiversity over time (S. Buckland et al. 2005; S. T. Buckland et al. 2011).

Arithmetic mean relative abundance. Arithmetic mean relative abundance is the average abundance of a species in a given site r relative to an abundance of that species at a fixed time point. It can be expressed as:

$$M_r = \frac{1}{S} \sum_{i \in S_r} \frac{a_{i,r,t}}{a_{i,r,1}} \tag{6}$$

Where $a_{i,r,t}$ is the abundance or count of individuals of species i in site r at time t and $a_{i,r,1}$ is the abundance of species i in site r at a fixed time point 1, which is usually the first time point (S. Buckland et al. 2005).

Geometric mean relative abundance. Geometric mean relative abundance is the nth root of the product of the abundances of n species in a given site r relative to their abundance at a fixed time point and can be expressed as:

$$G_r = \left(\prod_{i \in S_r} \frac{a_{i,r,t}}{a_{i,r,1}} \right)^{\frac{1}{S}} \tag{7}$$

This is generally expressed in log space as:

$$G_r = \exp\left(\frac{1}{S} \sum_{i \in S_r} \ln\left(\frac{a_{i,r,t}}{a_{i,r,1}} \right) \right) \tag{8}$$

Where $a_{i,r,t}$ is the abundance or count of individuals of species i in site r at time t and $a_{i,r,1}$ is the abundance of species i in site r at a fixed time point 1, which is usually the first time point (S. Buckland et al. 2005; S. T. Buckland et al. 2011).

These metrics each have their own benefits and limitations. Species richness is the most commonly used metric because presence absence data is the most common type of biodiversity data and it is easy to calculate (Özkan Tümer et al. 2026). However, species richness does not account for the abundance of species and can be biased towards rare species (Ferrier and Guisan 2006). Absolute abundance and relative abundance provide more information about the abundance of species in a given area, but they can be biased towards common species (Beery et al. 2021). The Shannon index combines species richness and evenness into a single metric and can provide a more comprehensive measure of biodiversity, but it can be difficult to interpret (Dušek and Popelková 2017). Arithmetic mean relative abundance and geometric mean abundance are useful for measuring changes in abundance over time and are sensitive to detection error (S. Buckland et al. 2005; Lamb et al. 2009; S. T. Buckland et al. 2011). It is important to look at a wide range of metrics when measuring biodiversity as each metric measures different aspects of biodiversity and responds differently to changes in biodiversity (Lamb et al. 2009).

2.2 Spatial Biodiversity Modeling

3 Methods/setup

3.1 Pipeline for Community Level Modeling

3.2 Data

3.3 Models

4 Results/ Case Study

5 Discussion

5.1 Findings

5.2 Ethical Discussion

6 Future Work/ Applications

Future Work includes a full comprehensive benchmark of community level models across a wide range of datasets and models. The integration with GBIF should should allow for easy facilitation of this. This would allow for a more systematic evaluation of community level models and provide insights into their strengths and weaknesses in predicting biodiversity patterns and trends. Furthermore, a comprehensive benchmark would allow for a more nuanced evaluation of community-level models and their applicability across different ecological contexts, including comparisons across taxonomic groups, spatial scales, and environmental predictor sets. This would ultimately contribute to the advancement of biodiversity modeling as a whole and inform conservation strategies and policy decisions.

7 Conclusion

A Appendix A: Github Repository

Includes the instructions there

B Appendix C: Additional Results

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